

The Hilbert Curve

A *space-filling curve* is a 1D line that curves around until it completely fills a 2D space without crossing over itself. German mathematician David Hilbert described his space-filling *Hilbert curve* in 1891. If you split a 2D area into a grid, the single, 1D line of the Hilbert curve can run through every cell in the grid.

[Figure 9-17](#) contains the first three recursions of the Hilbert curve. The next recursion contains four copies of the previous recursion, and the dashed line shows how the four copies connect to one another.

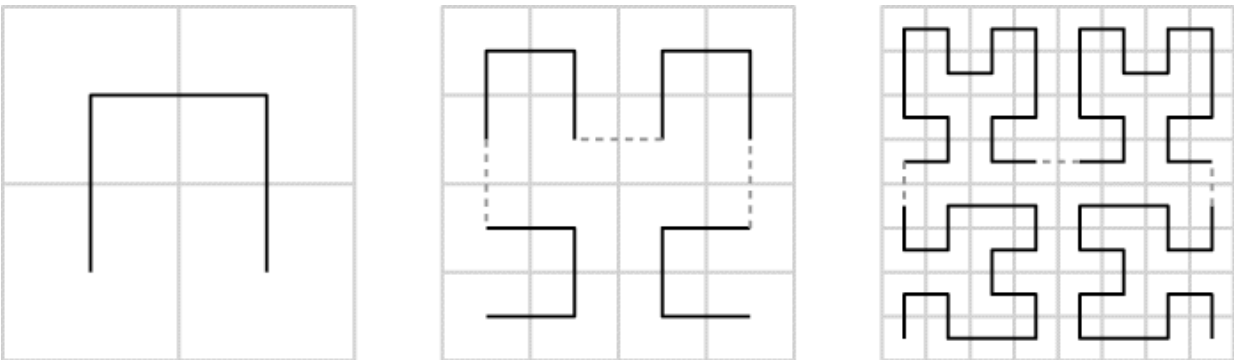


Figure 9-17: The first three recursions of the Hilbert space-filling curve

As the cells become infinitesimal points, the 1D curve can fill the entire 2D space the same way a 2D square does. Counterintuitively, this creates a 2D shape from a strictly 1D line!

The source code for *hilbertCurve.py* is as follows:

Python

```
import turtle
turtle.tracer(10, 0) # Increase the first argument to speed up the drawing.
turtle.setworldcoordinates(0, 0, 700, 700)
turtle.hideturtle()

LINE_LENGTH = 5 # Try changing the line length by a little.
ANGLE = 90 # Try changing the turning angle by a few degrees.
LEVELS = 6 # Try changing the recursive level by a little.
DRAW_SOLID = False
```